

1 Continuation Record: Franka Cube-Lift Attempts

1.1 Purpose and exact claim boundary

This section records the complete continuation state of the NVIDIA manipulation programme as of 8 August 2026. Its purpose is operational as well as scientific: a future development cycle should be able to resume from this point without reconstructing the controller history, repeating rejected experiments, or confusing contact evidence with a completed lift.

The current direct precision controller has *not* completed the strict cube-lift task. The independent success authority is a post-action cube height strictly greater than 0.10 m. The final fresh-world trial scored 0/1, and no controller from this continuation cycle was promoted. The programme nevertheless established four narrower physical capabilities:

1. the Franka can be brought safely into the correct object region by a bounded direct controller;
2. table and self-contact can be excluded from grasp evidence;
3. genuine object-only opposed fingertip contact can be produced; and
4. one sealed trajectory produced a genuine 7.9 mm physical cube micro-lift.

These are enabling results, not substitutes for the required lift. A historical protected controller also retains two genuine PhysX lifts in twelve trials, which proves that the robot, environment, and lift authority admit the task. It does not prove that the newer direct controller is reliable.

All paid experiments described here used NVIDIA Isaac Lab and PhysX on an NVIDIA L40S. NVIDIA Cosmos was not yet used to train or evaluate the grasp controller. Cosmos-based representation or world-model training remains a later stage, to begin only after a reproducible cube-lift baseline exists.

1.2 Runtime authority and success contract

The deployed proposal policy was deliberately denied privileged simulator state. At runtime it could use RGB, bounded robot proprioception, the episode step, and object-filtered fingertip touch. It could not use the true cube position, reward, teacher action, private contact geometry, or the evaluator’s success bit. True post-action cube position and policy mode were recorded privately for diagnosis after execution; they were never returned to the policy as input.

The causal task contract was

$$\begin{aligned} \text{safe approach} &\rightarrow \text{object-only opposed contact} \rightarrow \text{secure closure} \\ &\rightarrow \text{attachment evidence} \rightarrow \text{verification lift} \\ &\rightarrow h_{\text{cube}} > 0.10 \text{ m.} \end{aligned} \tag{1}$$

The controller was required to fail closed. One-finger touch, table contact, high simulator reward, commanded upward hand motion, or an unverified grasp could not be counted as a lift. This prevented the system from learning from false-positive contact or promoting a controller merely because the hand moved upward while the cube remained on the table.

1.3 Why the evaluation design was changed

Early matched searches revealed a simulator-authority problem. A retained 15 mm search controller changed from 2/6 to 1/6 under the same nominal seed, image, GPU, environment, and frozen policy. A deterministic audit seeded Python, NumPy, Torch, CUDA, and Warp; requested deterministic RTX; enabled enhanced PhysX determinism; and compared initial observations plus complete observation, action, and outcome chains. The first reset image was reconstructed exactly, but later reset images and causal chains diverged.

The likely boundary was sequential world reuse: later arms or episodes inherited solver, rendering, or contact history despite reset calls. Small 1/12 versus 2/12 comparisons were therefore too noisy to guide paid development. Subsequent work used one bounded fresh-world episode at a time

and treated a nominal seed as insufficient proof of identical late-contact conditions. Future tournaments must use either a newly reconstructed world for every arm and episode or a precommitted replicated statistical design with confidence bounds.

1.4 Failure-driven controller evolution

Table 1 records the important faults and the retained corrections. The sequence matters: several apparently sensible tuning attempts were invalidated by deeper action-surface or authority errors.

Table 1: Failure-driven evolution of the direct Franka controller.

Stage	Observed fault	Retained correction or conclusion
Premature closure	The gripper could begin closing while still 32–35 mm from the grasp point.	Enforce approximately 8 mm open-hand alignment, 5 mm contact alignment, and at most 3 mm closed-loop contact movements.
Reset authority	Identical nominal seeds did not reconstruct later trajectories.	Treat each arm/episode as an independent world; do not use tiny sequential comparisons as a reliable promotion authority.
Calibration	The initial contact geometry placed the gripper approximately 32 mm too low.	Correct the hand-to-grasp calibration; this produced the first genuine two-finger PhysX contact in the precision probes.
Action encoding	Direct joint targets were falsely clipped as though Isaac Lab exposed a normalized $[-1, 1]$ action surface.	Map absolute joint targets through the configured action scale and default offset while retaining Franka limits and a 35 mrad per-frame IK bound.
Retry semantics	Retry branches emitted seven zero arm actions; in this task, zero means the robot’s default pose rather than “hold current pose.”	Every reopen or retry now emits an absolute current-joint hold.
Tactile direction	The original lateral correction moved in the wrong direction, and repeated corrections could stack before the arm settled.	Use the empirically corrected direction, an execution cooldown, bounded cumulative movement, and later a symmetric bracket about the original approach.
Close timing	The close window ended before balanced force persisted for the four-frame attachment signature.	Extend the bounded close window and allow a small tactile force-balancing servo while preserving safety limits.
Attachment retention	A grasp signature was incorrectly treated as a force condition that had to remain invariant during upward acceleration.	Let the earned grasp latch persist while both tactile sensors retain contact; use the signature to authorize lift entry, not to demand stationary forces forever.
Visual occlusion	Close-range occlusion could move the remembered object target.	Ground the target while safely above the object and freeze the committed visual belief during the final approach.

Stage	Observed fault	Retained correction or conclusion
False contact	Table or robot self-contact could be interpreted as grasp evidence.	Bind fingertip sensing to the active object prim; only object-only left/right evidence can authorize closure or attachment.
Coarse controller	The learned coarse controller approached at roughly 5.9rad s^{-1} , trapped the wrist near the table, and left later controllers trying to recover from a collision-pinned state.	Remove it from the direct probe. A smooth, bounded direct trajectory owns motion from the initial safe pose onward.
Compliance and sag	The hand could sag by approximately 18 cm under load.	Increase damping, require settled motion before phase changes, and use measured Cartesian residual compensation; the observed sag was reduced to approximately 6 cm.
Lift transition	Older code could return to a less stable fixed joint trajectory after grasp verification.	Route approach, contact, verification, and full-lift phases through the settled Cartesian controller.

1.5 Experimental evidence ladder

The progression was not monotonic because fresh PhysX worlds produced different late-contact states. Table 2 separates the strongest evidence at each level.

Table 2: Strongest retained evidence from the continuation cycle.

Evidence level	Best observation	Interpretation
Historical task completion	2/12 protected PhysX lifts	The environment and lift pathway are physically feasible, but historical reliability is low.
Broad opposed contact	88 opposed-contact frames in an instrumented probe	Both fingers can reach the cube; alternating force prevented the sealed four-frame attachment signature.
Physical cube motion	$0.0210 \rightarrow 0.0289\text{ m}$	Genuine 7.9 mm micro-lift with median fingertip forces of approximately 67 N and 79 N; below the task threshold.
Object-filtered stable contact	19 consecutive opposed-contact frames	Best evidence under the hardened final architecture; finger width was approximately 23.1 mm and forces were about 41 N and 33 N.
Final bounded attempt	0/1; three one-sided frames; no closure	The bracketed controller failed to reacquire contact after its first causal correction and correctly refused an unverified lift.

The object-filtered contact run failed the deliberately strict attachment signature because its force-balance ratio was approximately 0.80, below the required 0.85. It subsequently reopened after the bounded close window. This run proves that secure-looking grasp geometry is achievable, but it does not establish repeatable attachment.

The micro-lift run exposed a different boundary. Its strongest grasp window reached a force-balance ratio of 0.857, but the measured finger width was 28.26 mm, only 0.26 mm beyond the probe’s then-current maximum. A probe-only 3 mm width margin was added, while the default promoted controller retained its stricter sealed threshold. A fresh repeat did not reacquire contact, so the relaxed probe was not credited with a full lift.

1.6 Final attempt and terminal state

The last authorized paid attempt used a 350-step fresh world with symmetric tactile bracketing. If the first lateral correction failed, a second correction was prepared to test the opposite side of the original approach rather than continuing farther in the same direction. Verified grasp and lift phases were also assigned to the stable Cartesian controller.

The sealed outcome was:

$$\begin{aligned} N_{\text{lift}} &= 0/1, & h_{\text{start}} &= 0.052057 \text{ m}, \\ h_{\text{max}} &= 0.052057 \text{ m}, & \Delta h_{\text{cube}} &= 0, \\ N_{\text{dual}} &= 0, & N_{\text{one-sided}} &= 3, \\ N_{\text{unsafe}} &= 0. \end{aligned}$$

The controller observed persistent one-sided contact, issued one causal reopen and re-centre action, and then failed to regain contact. The second side of the bracket was never reached because persistent tactile evidence did not recur. The fingers never closed, no attachment signature was earned, and no lift command was authorized. This is a safe and correctly rejected episode.

All 33 focused controller, tactile, and evaluator-contract tests passed before the final run. The trajectory archive contains 350 RGB frames, bounded proprioception, actions, fingertip touch, rewards for audit only, true post-action cube position for private evaluation, and the per-step policy mode. Both NVIDIA workspaces were stopped after evidence retrieval. No additional paid retry was launched, no new controller was promoted, and no Git commit was made for the experimental working state.

1.7 Current diagnosis

The principal blocker is **repeatable acquisition of a balanced two-finger grasp in a fresh world**. It is no longer accurate to describe the problem as simply “the arm cannot reach the cube” or “the lift command is missing.” The arm has reached the object, generated opposed contact, closed around it, and physically begun lifting it. What remains unreliable is the last-centimetre sequence that converts approach into the same balanced contact state on demand.

Three effects must not be conflated:

1. **world variance:** nominally identical resets can diverge after contact history accumulates;
2. **residual Cartesian error and compliance:** the commanded target is not necessarily the measured loaded hand pose; and
3. **discrete tactile correction:** a one-sided signal can disappear after a large re-centering move before the controller gathers enough evidence to test the opposite side.

The upward Cartesian path is prepared but remains weakly tested because grasp acquisition rarely authorizes it. The immediate research problem is therefore to make attachment entry repeatable, then test whether attachment survives a small vertical displacement before attempting the full task lift.

1.8 Required restart protocol

When this work resumes, the following order should be preserved.

1. **Begin from the retained evidence, not a broad search.** Reanalyse the 19-frame object-filtered opposed-contact window and the 7.9 mm micro-lift. Record measured hand pose, finger width, left/right force, velocity, and policy phase at entry, peak balance, closure, and separation.

2. **Reconstruct a fresh simulator world for every episode.** Record the initial RGB and bounded-observation hashes. Never compare arms sequentially in one reused world and call the result matched.
3. **Replace the large discrete re-centering jump with a bounded local contact search.** Around the final measured contact pose, test a precommitted symmetric set of millimetre-scale lateral offsets. Each move must settle before new touch evidence is interpreted. The search must use fingertip touch and proprioception only; true cube pose remains evaluator-owned.
4. **Separate acquisition from attachment qualification.** First require repeatable opposed object-only contact and a realistic finger-width band. Then close gradually while the tactile balance servo minimizes left/right force asymmetry. Log why each candidate window fails the signature rather than changing several thresholds at once.
5. **Use a 3–4 cm verification lift.** Once attachment is earned, command a slow Cartesian rise while holding closure. Attachment is verified only if the cube follows the hand. If it slips, capture the exact force, width, hand acceleration, and object-height frame at separation.
6. **Attempt the full lift only after verification.** Continue beyond the independent 0.10 m cube-height threshold. Simulator reward must remain diagnostic and cannot replace this authority.
7. **Demand replication before promotion.** A single lift is a development breakthrough, not a reliable skill. The initial reliability target should be at least 4/6 safe lifts in independently reconstructed worlds, followed by held-out seeds.

Each paid cycle should change one causal mechanism, run the smallest experiment that can falsify it, retrieve immutable evidence, and stop both GPUs. If a trial does not reach object-only contact, attachment thresholds are irrelevant; if it reaches balanced attachment but slips during the verification lift, the next change belongs to closure, friction, wrist stability, or acceleration— not approach geometry.

1.9 Path from one cube to general grasp competence

The present controller is object-generic only in a limited architectural sense: contact filtering binds to the active object prim rather than a cube-specific collision path, and the runtime contract does not expose the cube pose. Empirical competence has been demonstrated only on the cube.

After the cube gate is repeatable, the curriculum should expand one factor at a time across object width, height, aspect ratio, mass, friction, orientation, occlusion, and initial lateral offset. Training should distinguish at least four reusable skills:

visual grounding, pre-grasp placement, tactile acquisition, load-preserving transport. (2)

Only after transfer succeeds across held-out objects should NVIDIA Cosmos be introduced for predictive visual or world-model learning. Cosmos should be evaluated as a challenger representation under the same physical authority; it must not receive private cube state or convert imagined success into competence credit.

1.10 Continuation artifact manifest

The following files define the restart point. Paths are absolute because the working programme was executed from the COMDEX repository on the development host.

Purpose	Artifact
Final controller	/Users/kevinrobinson/dev/COMDEX/integrations/isaac_lab/aion_precision_franka_policy.py
Runner and private diagnostic capture	/Users/kevinrobinson/dev/COMDEX/integrations/isaac_lab/aion_isaac_lab_runner.py
Direct probe configuration	/Users/kevinrobinson/dev/COMDEX/integrations/isaac_lab/physx_direct_joint_ik_contact_probe.py

Purpose	Artifact
Determinism and precision-contact diagnosis	<code>/Users/kevinrobinson/dev/COMDEX/docs/aion/AION_PHYSX_DETERMINISM_AND_PRECISION_CONTACT_REPORT.md</code>
Fresh-world direct-servo and micro-lift report	<code>/Users/kevinrobinson/dev/COMDEX/docs/aion/AION_FRESH_WORLD_DIRECT_JOINT_SERVO_2026-08-08.md</code>
Object-filtered architecture report	<code>/Users/kevinrobinson/dev/COMDEX/docs/aion/AION_OBJECT_FILTERED_GENERAL_GRASP_CONTROL_LEAP_2026-08-08.md</code>
Final bounded-attempt report	<code>/Users/kevinrobinson/dev/COMDEX/docs/aion/AION_FINAL_BRACKETED_CUBE_LIFT_ATTEMPT_2026-08-08.md</code>
Micro-lift receipt	<code>/Users/kevinrobinson/dev/COMDEX/results/immutable/isaac_precision_franka/direct_joint_opposed_cooldown_20260808_v1.json</code>
Micro-lift trajectory	<code>/Users/kevinrobinson/dev/COMDEX/results/immutable/isaac_precision_franka/direct_joint_opposed_cooldown_20260808_v1.npz</code>
Best final-architecture opposed-contact receipt	<code>/Users/kevinrobinson/dev/COMDEX/results/immutable/isaac_precision_franka/object_filtered_phase2_cartesian_20260808_v1.json</code>
Best final-architecture opposed-contact trajectory	<code>/Users/kevinrobinson/dev/COMDEX/results/immutable/isaac_precision_franka/object_filtered_phase2_cartesian_20260808_v1.npz</code>
Final attempt receipt	<code>/Users/kevinrobinson/dev/COMDEX/results/immutable/isaac_precision_franka/object_filtered_final_bracketed_cartesian_lift350_20260808_v1.json</code>
Final attempt trajectory	<code>/Users/kevinrobinson/dev/COMDEX/results/immutable/isaac_precision_franka/object_filtered_final_bracketed_cartesian_lift350_20260808_v1.npz</code>
Focused controller tests	<code>/Users/kevinrobinson/dev/COMDEX/backend/tests/test_precision_franka_adapter.py</code>
Focused tactile tests	<code>/Users/kevinrobinson/dev/COMDEX/backend/tests/test_active_tactile_franka_adapter.py</code>
Runtime authority/filter tests	<code>/Users/kevinrobinson/dev/COMDEX/backend/tests/test_isaac_contract_filters.py</code>

1.11 Handover verdict

The cube-lift task remains open. The programme progressed from unstable approach and false contact to object-filtered opposed contact, a genuine 7.9 mm micro-lift, stable 19-frame grasp geometry, and a fail-closed Cartesian lift architecture. It did not convert those advances into a fresh post-action cube height above 0.10 m before the bounded development cycle ended.

The shortest credible route on resumption is not another broad policy search. It is a fresh-world, millimetre-scale acquisition experiment around the proven opposed-contact pose, followed by a slow 3–4 cm attachment test and then the full lift. Until that sequence is replicated, AION should be described as having learned important components of grasping without yet mastering reliable cube pickup.